

Burmese Automatic Speech Recognition Using Kaldi

From GMM-HMM to Convolutional-LSTM Acoustic Modelling

Course / Assignment : AIEF - 2 , Assignment 4

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The image shows a digital form titled "Project Report" for "Burmese Automatic Speech Recognition (Kaldi)". The form is divided into two main columns: "GMM-HMM" and "NEURAL NETWORK". Each column has fields for "Model name", "Model type", and "Key notes". Below these is a "PERFORMANCE COMPARISON (HIGHER IS BETTER)" section featuring a horizontal bar chart comparing "Model A" (blue) and "Model B" (yellow) across four metrics: Accuracy, Precision, Recall, and F1-score. The x-axis represents the "Score" from 0.00 to 1.00. At the bottom, there is a "Decision notes" section with three numbered lines for input. The form is presented on a white background with a dark blue border, and the overall design is clean and professional.

Project Report

Burmese Automatic Speech Recognition (Kaldi)

GMM-HMM

Model name

Model type

Key notes

NEURAL NETWORK

Model name

Model type

Key notes

PERFORMANCE COMPARISON (HIGHER IS BETTER)

	Model A	Model B
Accuracy	0.75	0.65
Precision	0.65	0.85
Recall	0.60	0.75
F1-score	0.65	0.75

Score: 0.00, 0.25, 0.50, 0.75, 1.00

Decision notes

-
-
-

WHAT IS AUTOMATIC SPEECH RECOGNITION?

A	ASR converts spoken language → text
	Main components: ○ Acoustic Model ○ Pronunciation Lexicon ○ Language Model ○ Decoder
	Our project focuses on Burmese ASR
	We implemented both: ○ Traditional GMM-HMM ○ Neural DNN-based approaches

PROJECT OBJECTIVES

1. Collect and prepare a Burmese speech dataset
2. Build a Kaldi-based ASR pipeline
3. Extract MFCC features
4. Create a Burmese pronunciation lexicon
5. Build Unigram, Bigram and Trigram LMs
6. Train GMM-HMM acoustic models
7. Test speaker adaptation with fMLLR
8. Experiment with DNN-based acoustic models
9. Compare all systems using WER and SER
10. Identify the best-performing model

DATA COLLECTION

Data Collection Process

Prepare Burmese Sentences



Assign Sentences to Speakers



Speakers Record Their Voices



Collect Audio Recordings



Match Audio with Transcriptions



Organize Kaldi Data Files

Dataset Characteristics

Speakers' Information

- Male : 4 ယောက်
- Female : 3 ယောက်

WAV audio recordings: 30

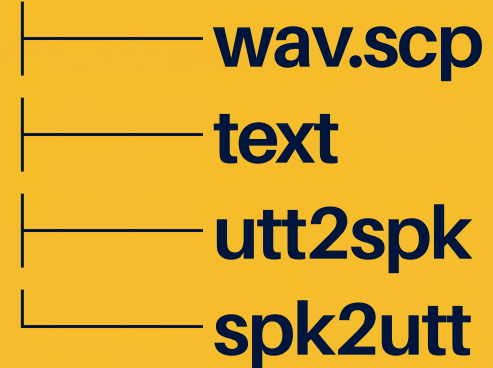
- 5,250 recorded sentences/utterances

Approximately 4-5 seconds per utterance

We collected speech recordings from multiple speakers so that the system could learn the relationship between Burmese speech sounds and their corresponding text.

KALDI DATA FILES

data/train/



A diagram showing four files: wav.scp, text, utt2spk, and spk2utt. Each file name is preceded by a horizontal line, and these lines are connected by vertical lines on the left, forming a bracket-like structure.

- wav.scp
- text
- utt2spk
- spk2utt

data/test/



A diagram showing four files: wav.scp, text, utt2spk, and spk2utt. Each file name is preceded by a horizontal line, and these lines are connected by vertical lines on the left, forming a bracket-like structure.

- wav.scp
- text
- utt2spk
- spk2utt

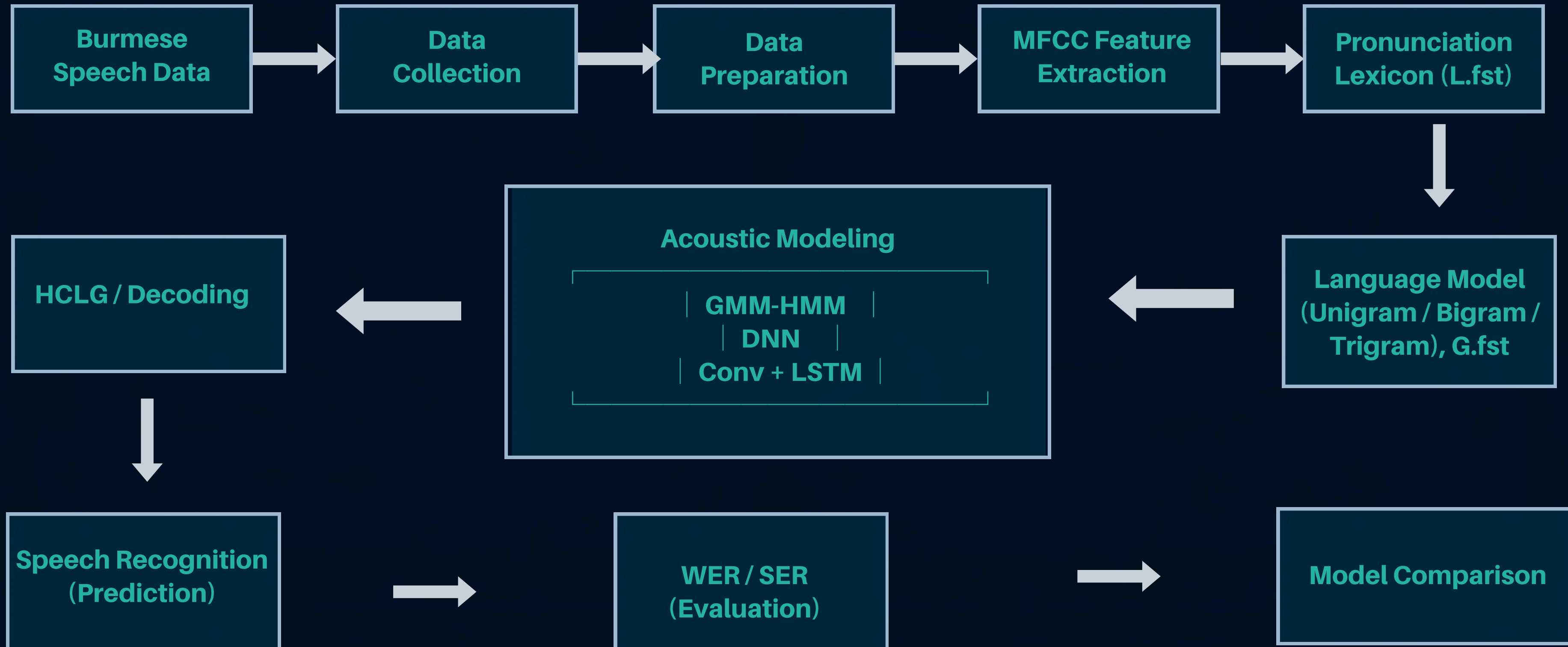
Processing

- Checked audio files
- Removed missing recordings
- Duplicated Data
- Matched audio with transcripts
- Created speaker mappings
- Validated Kaldi data directories

FINAL DATASET:

2,999 TRAINING + 1,502 TESTING UTTERANCES

OVERALL SYSTEM PIPELINE



ACOUSTIC FEATURES

- Sampling frequency: 16 kHz
- 23 Mel filter banks
- 13 MFCC coefficients
- CMVN applied
- High-resolution features were also prepared for DNN experiments

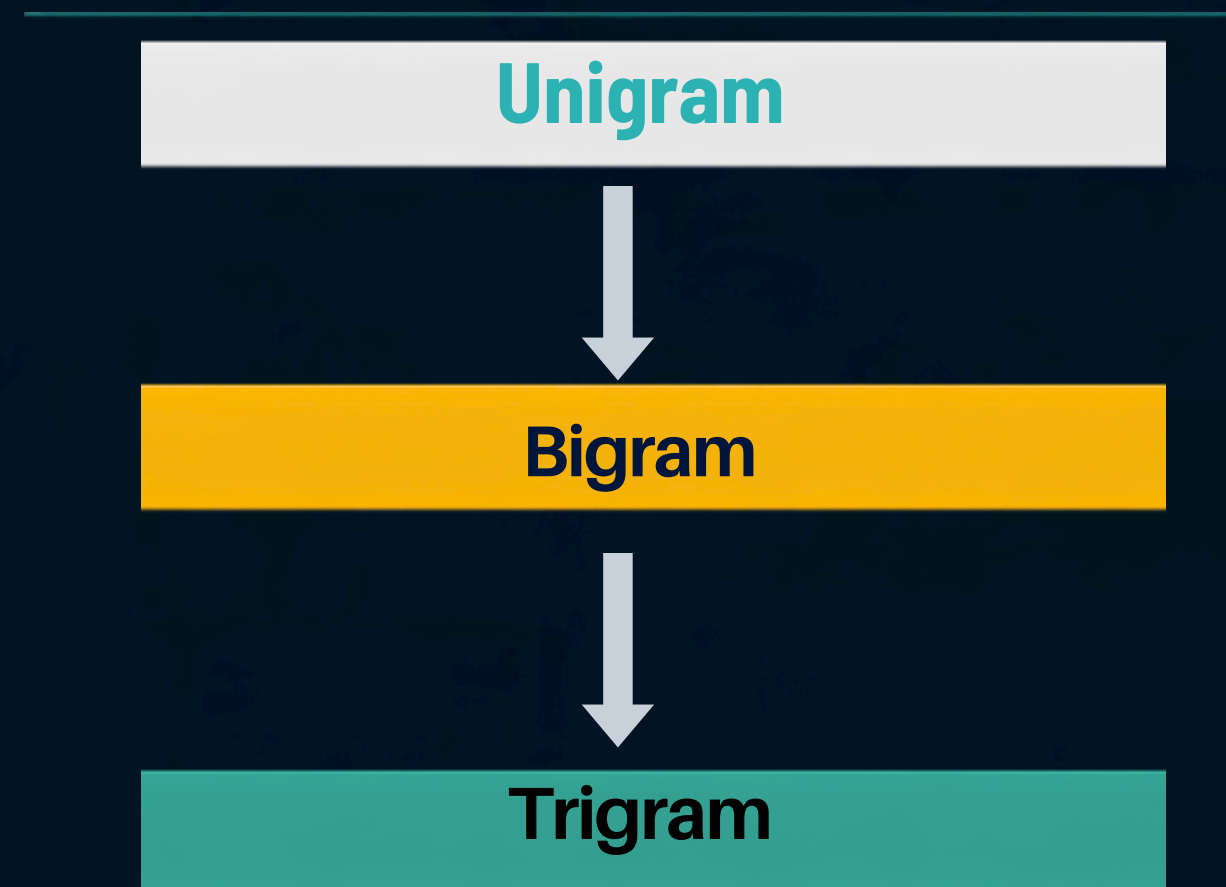
WHY MFCC?

MFCC features represent important characteristics of human speech and are widely used as acoustic features in speech recognition.

PRONUNCIATION LEXICON (L.FST)

- Character-based Burmese phone representation
- 48 Burmese character units + SPN
- Silence phone: SIL
- Vocabulary: 136 words

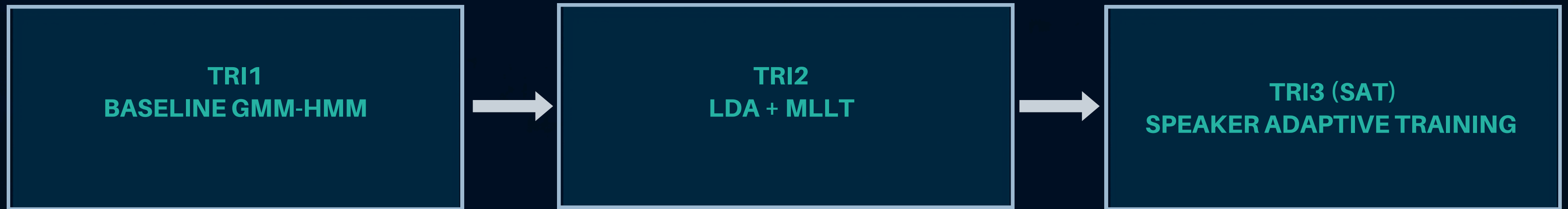
Language Models (G.fst)



The language model helps the decoder determine which word sequences are more likely.

TRADITIONAL ACOUSTIC MODELS

We gradually increased the complexity of the GMM-HMM system.



Then we tested:

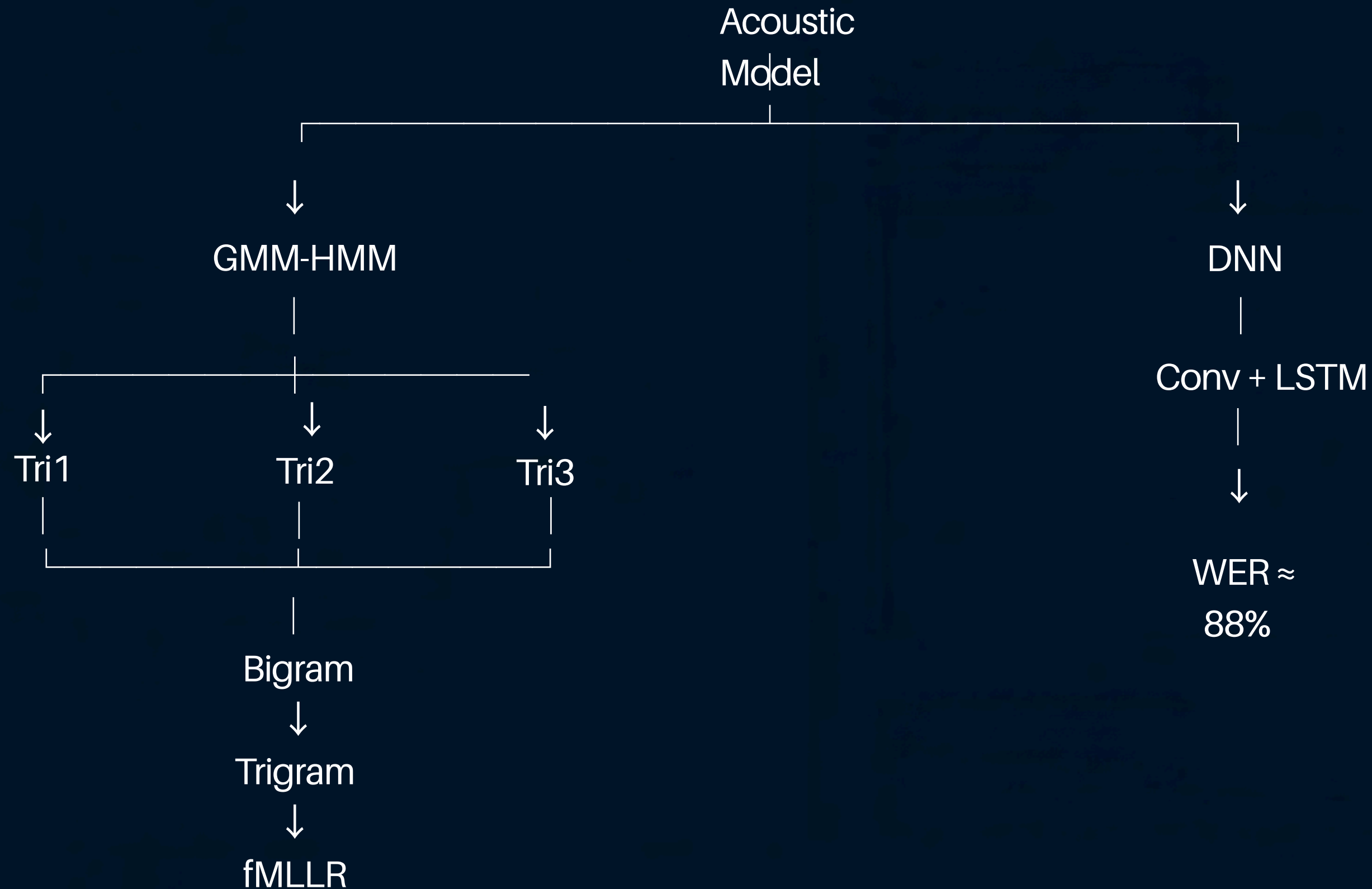
Tri3 + fMLLR decoding

This gives a clear experimental story:

Tri1 → Tri2 → Tri3 → fMLLR

EXPERIMENTAL STRATEGY

We tested the systems progressively:



Testing principle:

- Same test set
- Same reference transcripts
- Decode each system
- Calculate WER/SER
- Compare error types

GMM-HMM RESULTS

System	WER	SER
Tri1 + Bigram	124.46%	92.01%
Tri2 + Bigram (LDA+MLLT)	103.57%	95.14%
Tri3 + Bigram (SAT)	96.92%	95.12%
Tri3 + Trigram	96.06%	94.87%
Tri3 + fMLLR + Trigram	104.39%	92.48%

Observation
Tri3 + Trigram was the best traditional GMM-HMM configuration.

Error Analysis: Substitution, Deletion & Insertion

System	Substitution	Insertion	Deletion
Tri1 + Bigram	1,978	1,852	531
Tri2 + Bigram (LDA+MLLT)	1,738	666	1,225
Tri3+ Bigram (SAT)	1,563	1,339	494
Tri3 + Trigram	1,554	1,345	467
Tri3 + fMLLR + Trigram	1,800	956	902

Evaluation

Substitution Errors

A word in the hypothesis is replaced with a different word from the reference

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ

ခေါင်းကိုက်

နေ

သည်

Deletion Errors

A word from the reference is missing in the hypothesis

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ



နေ

သည်

Insertion Errors

An extra word appears in the hypothesis that is not in the reference

Reference

လူနာ

မှာ

ဖျားနာ

နေ

သည်

Hypothesis

လူနာ

မှာ

အတော်လေး

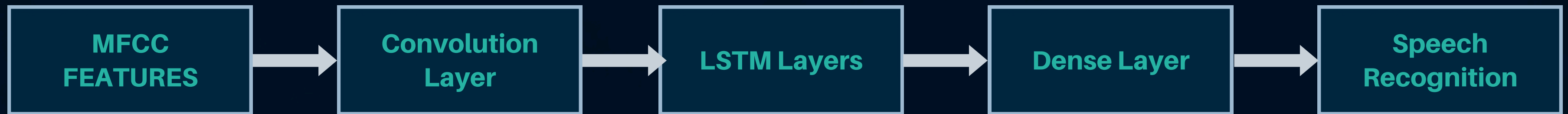
ဖျားနာ

နေ

သည်

DNN ACOUSTIC MODEL

Convolution + LSTM



Why DNN?

Traditional GMM-HMM models have limitations in learning complex speech patterns.

DNNs can learn more complex nonlinear relationships between:

Acoustic Features → Speech Units

DNN RESULTS

WER ■ Tri3 + Trigram ■ Conv + LSTM

96.06%

72.6%

SER ■ Tri3 + Trigram ■ Conv + LSTM

94.87%

84.62%

The DNN system reduced WER by approximately 24 **percentage** points compared with the best completed GMM-HMM system.

OVERALL MODEL COMPARISON

Model	WER	Key Observation
Tri1 + Bigram	124.46%	Baseline
Tri2 + Bigram	103.57%	LDA + MLLT improved performance
Tri3 + Bigram	96.92%	SAT further improved WER
Tri3 + Trigram	96.06%	Best GMM-HMM
Tri3 + fMLLR + Trigram	104.39%	Performance decreased
Conv + LSTM DNN	72.6%	Best Overall

ANALYSIS & FINDINGS

Finding 1 - Acoustic Modeling

Tri1 → Tri2 → Tri3 consistently improved WER.
124.46% → 103.57% → 96.92% → 96.06%

Finding 2 - Language Modeling

Bigram provided a major improvement compared with the unigram baseline.
Trigram gave only a small additional improvement.

Finding 3 - fMLLR

- Reduced deletion errors
 - Increased substitution errors
 - Increased insertion errors
 - Overall WER became worse
-

Finding 4 - DNN

Conv + LSTM achieved 72.6% WER, outperforming the traditional GMM-HMM systems.

CHALLENGES & FUTURE WORK

CHALLENGES

- Still Learning & Improving...
- Not sufficient memory (RAM)
- Limited amount of Burmese speech data
- Limited vocabulary
- Burmese linguistic/Unicode complexity
- High recognition error rate

FUTURE WORK

- Explore LSTM approach with Kaldi
- Collect more Burmese speech data
- Increase speaker and speaking-style diversity
- Improve pronunciation lexicon
- Improve language-model smoothing
- Experiment with stronger CNN/LSTM architectures

CONCLUSION

We successfully developed and evaluated a **Burmese ASR system** using Kaldi. We compared traditional GMM-HMM models with neural acoustic modeling. The best traditional system, Tri3 + Trigram, achieved **96.06% WER**, while the Convolution + LSTM DNN achieved approximately **72% WER**, giving the best overall performance in our experiments.

REFERENCES

- Povey, D., Ghoshal, A., Boulianne, G., et al. (2011). The Kaldi Speech Recognition Toolkit. IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU).
- Kaldi Speech Recognition Toolkit Documentation. Kaldi ASR Project.
- Our Project Dataset, Experiments, and Evaluation Results. Group 4, Assignment 4 — ASR Mini Project, AIE-F Batch-2, 2026.
- Recording Tool: [ye-kyaw-thu/AIE-F-B2/tree/main/assignment/assignment-4/recording_tool](#)
- LDA & MLLT Documentation
- Towards Speaker Adaptive Training of Deep Neural Network Acoustic Models

THANK YOU!